1.(a)	$\frac{dy}{dx} = \frac{1}{8} \times 3x^2 - 24 \times -\frac{1}{2} x^{-\frac{3}{2}}$	M1
	$\frac{dy}{dx} = \frac{3}{8} x^2 + 12x^{-\frac{3}{2}}$	A1 A1 **(3)**
(b)	$\left.\frac{dy}{dx}\right\|_{x=4} = \frac{3}{8} \times 4^2 + 12 \times 4^{-\frac{3}{2}} = (7.5)$	M1
	$y + 3 = 7.5(x - 4) \Rightarrow y = 7.5x - 33$	M1 A1 **(3)** **(6 marks)**

2.(a)	$\frac{1}{4-2\sqrt{2}} = \frac{1}{4-2\sqrt{2}} \times \frac{4+2\sqrt{2}}{4+2\sqrt{2}}$	M1
	$= \frac{4+2\sqrt{2}}{16-8} = \frac{1}{2} + \frac{1}{4}\sqrt{2}$ oe	A1
		(2)
(b)	$4x = 2\sqrt{2}x + 20\sqrt{2} \Rightarrow \left(4-2\sqrt{2}\right)x = 20\sqrt{2}$	M1
	$\Rightarrow x = \frac{20\sqrt{2}}{\left(4-2\sqrt{2}\right)} = 20\sqrt{2} \times (\text{a})$	dM1
	$\Rightarrow x = 20\sqrt{2} \times \left(\frac{1}{2} + \frac{1}{4}\sqrt{2}\right) = 10 + 10\sqrt{2}$	A1
		(3)
		(5 marks)

3.(a)	Attempts perimeter of garden = $2\times5x+2\times(6x-2)$	M1
	Sets $2\times5x+2\times(6x-2)>29\Rightarrow22x>33$	dM1
	$\Rightarrow x>\frac{33}{22}\Rightarrow x>1.5$ *	A1*
		(3)
(b)	Attempts area of garden = $2x(2x-1)+3x(6x-2)$	M1
	Sets $A<72\Rightarrow\ 22x^2-8x-72<0$	A1
	Finds critical values $11x^2-4x-36\Rightarrow x=-\frac{18}{11},2$	M1
	Chooses inside region	ddM1
	$-\frac{18}{11}<x<2$	A1
		(5)
(c)	$1.5<x<2$	B1
		(1)
		(9 marks)

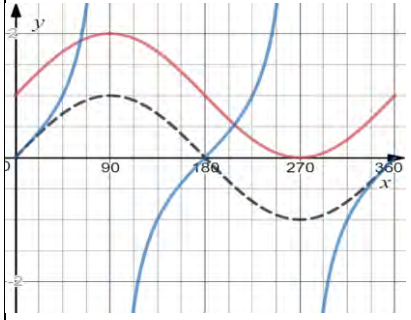
4.	$\frac{4x^2+1}{2\sqrt{x}} = \frac{4x^2}{2\sqrt{x}} + \frac{1}{2\sqrt{x}} = 2x^{\frac{3}{2}} + \frac{1}{2}x^{-\frac{1}{2}}$	M1 A1
	$\int \frac{4x^2+1}{2\sqrt{x}}\,dx = \frac{4}{5}x^{\frac{5}{2}} + x^{\frac{1}{2}} + c$	M1 A1 A1 **(5 marks)**

5.(a)	$2x^3+3x^2-35x=0 \Rightarrow x\left(2x^2+3x-35\right)=0$	M1
	$(2x-7)(x+5)=0 \Rightarrow x=...$	dM1
	$x=-5,\ 0,\ \frac{7}{2}$	A1
		(3)
(b)	$2(y-5)^6+3(y-5)^4-35(y-5)^2=0$	
	States that $y=5$ is a solution	B1
	$(y-5)^2="\frac{7}{2}" \Rightarrow y=...$	M1
	$y=5+\sqrt{\frac{7}{2}}$ or $y=5-\sqrt{\frac{7}{2}}$ or exact equiavlent	A1ft
	Both $y=5+\sqrt{\frac{7}{2}}$ and $y=5-\sqrt{\frac{7}{2}}$ or exact equivalent.	A1
		(4)
		(7 marks)

6.(a)	Sets $4x+c=x(x-3)$ and attempts to write as a 3TQ	M1
	Uses $b^2=4ac$ for their $x^2-7x-c=0$	dM1
	Correct equation $49=-4c$ or $49+4c=0$	A1
	$c=-12.25$ oe	A1
		(4)
(b)	Attempt to solve $x^2-7x-c=0$ with their c	M1
	Attempt to find the y coordinate for their x coordinate	dM1
	$\left(\frac{7}{2},\frac{7}{4}\right)$ oe	A1
		(3)
		(7 marks)

7.(a)	Attempts to use $\frac{1}{2}r^2\theta$ with $r=6$ and any allowable angle θ	M1
	Full method to find area $\frac{1}{2}\times 6^2\times(2\pi-0.7)$ or $\pi\times 6^2-\frac{1}{2}\times 6^2\times 0.7$	M1
	$=100.5\,\text{cm}^2$ (awrt)	A1
		(3)
(b)	Attempts $\frac{\sin\angle ADO}{6}=\frac{\sin 0.7}{5}\Rightarrow \sin\angle ADO=0.77...$	M1 A1
	$\angle ADO=2.258$ (awrt)	A1
		(3)
(c)	Attempts arc length $ABC=6\times(2\pi-0.7)$ 33.50	M1
	Attempts length OD $\frac{\sin(\pi-0.7-"2.258")}{OD}=\frac{\sin 0.7}{5}\Rightarrow OD=...$ 1.42	M1
	Full method to find perimeter = "33.50"+5+ 6−"1.42"	ddM1
	= 43.1 cm	A1
		(4)
		(10 marks)
Alt (c)	Alternative for arc length $ABC=12\pi-6\times 0.7$	M1
	Alternative for finding OD using the cosine rule $OD^2=6^2+5^2-2\times 6\times 5\cos(\pi-0.7-"2.258")\Rightarrow OD$	M1
	Solutions where candidate changes to degrees Look for angle AOD = awrt 40° to score M marks	
7.(a)	Attempts to use $\frac{\theta}{360}\pi r^2$ with $r=6$ and angle $\theta=$ awrt 40 or 320	M1
	Full method to find area $\frac{(360-\text{awrt }40)}{360}\times\pi 6^2$ or $\pi\times 6^2-\frac{\text{awrt }40}{360}\times\pi 6^2$	M1
	$=100.5\,\text{cm}^2$ (awrt)	A1
		(3)
(b)	Attempts $\frac{\sin\angle ADO}{6}=\frac{\sin 40°}{5}\Rightarrow \sin\angle ADO=0.77...$	M1 A1
	$\angle ADO=129.4°$ (awrt)	A1
		(3)
(c)	Attempts arc length $ABC=\frac{(360-40)}{360}\times 2\pi 6$ 33.50	M1
	Attempts length OD $\frac{\sin(180-40-"129.4")}{OD}=\frac{\sin 40}{5}\Rightarrow OD=...$ 1.42	M1
	Full method to find perimeter = "33.50"+5+ 6−"1.42"	ddM1
	= 43.1 cm	A1
		(4)
		(10 marks)

8.(a)	Substitutes $x=4$ in $\mathrm{f}'(4)=4\times2-2-\dfrac{8}{3\times4^2}=\left(\dfrac{35}{6}\right)$	M1
	Attempts to find the gradient of the perpendicular = $-\dfrac{6}{35}$	dM1
	Attempts the normal $y-1=-\dfrac{6}{35}\times(x-4)\Rightarrow 6x+35y-59=0$	M1A1
		(4)
(b)	$\mathrm{f}'(x)=4x^{\frac{1}{2}}-2-\dfrac{8}{3x^2}\Rightarrow \mathrm{f}(x)=\dfrac{8}{3}x^{\frac{3}{2}}-2x+\dfrac{8}{3x}(+c)$	M1 A1 A1
	$x=4,\mathrm{f}(x)=1\Rightarrow\ 1=\dfrac{8}{3}\times8-8+\dfrac{2}{3}+c\Rightarrow c=\ldots(-13)$	dM1
	$\mathrm{f}(x)=\dfrac{8}{3}x^{\frac{3}{2}}-2x+\dfrac{8}{3x}-13$	A1
		(5)
		(9 marks)

9. (a)	$(270°, -4)$		B1 B1 **(2)**
(b)		For $y = 1 + \sin\theta$	B1
		$y = \tan\theta$	B1 **(2)**
(c)	(i) $6 \times 2 = 12$ (ii) 11		M1 A1 B1 ft **(3)** **(7 marks)**

10 (a)	$P=\left(-\frac{1}{2},0\right)$	B1 **(1)**
(b)	$f(x)=(x-4)(2x+1)^2 \Rightarrow f(x)=ax^3+bx^2+cx+d$ $=4x^3-12x^2-15x-4$ oe $f'(x)=12x^2-24x-15$	M1 A1 dM1 A1 **(4)**
(c)	Attempts $f'(2.5)=12\times2.5^2-24\times2.5-15=0$ Finds y coordinate for $x=2.5$ $y=-54$	M1A1 A1 **(3)**
(d)	$a=-\frac{1}{2},(+)\,4$	B1, B1 **(2)** **(10 marks)**